METRIC

A-A-50626
18 July 1994

COMMERCIAL ITEM DESCRIPTION

SWITCH, DOOR TYPE IC/DM-1-( )

The General Services Administration has authorized the use of this commercial item description for all Federal agencies

1 Abstract. This Commercial Item Description covers requirements for door switches for security applications in naval ships Installations include hatch covers, watertight doors, and metal joiners

2 Salient characteristics

2 1 Integral components Integral components include switch unit, hardened steel strikers and a steel corrosion resistant base plate

2 2 Physical characteristics Switch dimensions and component orientation shall be as specified in Figures 1, 2, 3 and 4.

2 3 Enclosure. Switches shall be splashproof when mounted in a normal position. Liquid droplets or solid particles sprayed at the equipment at angles of no greater than ± 100 degrees from the vertical shall cause no harmful operational or functional effects

2 4 Corrosion resistance Door switches shall be corrosion resistant

2 5 Tamper resistance The switches shall be tamper-resistant

2 6 Switch configuration The switch functional configuration, including the pertinent electrical characteristics, shall be consistent with the data in table I or as otherwise specified in the contract or order

Beneficial comments, recommendations, additions, deletions, clarifications, etc and any other data which may improve this document should be sent by letter to Commander, Naval Sea Systems Command, ATTN SEA 03R42, 2531 Jefferson Davis Hwy , Arlington, VA 22242-5160

AMSC N/A FSC 5930

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TABLE I Electrical Features

Symbol	Switch	Contacts close on	Voltage	Current	
				AC[1]	DC
A	SPST	Rise	115	4 0 3 0	
C	SPST	Rise	115 230 440 500	7 5 5 0 3 0 -	4 6 2 3 1 2 1 0
D	SPST	Drop	115	4 0	3.0
F	SPST	Drop	115 230 440 500	7 5 5 0 3 0 -	4 6 2 3 1 2 1 0
G	SPDT	Rise	115	4 0	3 0
J	SPDT	Rise	115 230 440 500	7 5 5 0 3 0 -	4 6 2 3 1 2 1 0
K	SPDT	Drop	115	4 0	3 0
M	SPDT	Drop	115 230 440 500	7 5 5 0 3 0 -	4 6 2 3 1 2 1 0
O	SPST or SPDT	As required	120	0.5	0 5
P	SPST	Drop	1000	-	1 0
S	SPST	Rise	115	3 0	0 5
T	SPST	Drop	115	3 0	0 5
U	SPST	Rise	115	0 05	0 05
V	SPST	Drop	115	0 05	0 05
W	DPST	-	115	0 3	0 2

[1]Alternating current (ac), shall be 60 (Hz)

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2 7 Mechanical range features Travel range of the type IC/DM-1-( ) door switch roller shall be 48 mm minimum Full travel of the roller shall not be injurious to the door switch mechanism Dead band requirements shall be as specified in Figure 6

2 8 Temperature Door switches shall operate continuously throughout the temperature range of -20° C to 50° C

2 9 Mounting Door switches shall be mountable in accordance with Figure 1 The entire inner switch assembly shall be rotatable 90 degrees with respect to the switch strike plate and shall be adjustable to desired position with respect to flat top surface as specified in Figure 8.

2 10 Dimensions Dimensions for mounting shall be as shown in Figure 1

2 11 Identification plate An identification plate, as specified in Figure 5, shall be provided

2 12 Instruction plate An instruction plate suitable for mounting on the inside of the front cover shall be provided The plate shall be as specified in Figure 6 and shall include

a Wiring Diagram
b Roller Adjustment
c Dead band upper limit
d Dead band lower limit

3 Quality assurance provisions

3 1 Identification numbers Name of the switch and type designation and the manufacturer's name and part number shall be shown on an identification plate.

3 2 Ordering data Acquisition documents must specify the following

a Title, number, and date of this CID
b Type and configuration of switch required
c Types of washers, grommet and caps when required, and whether both halves of connector shall be provided

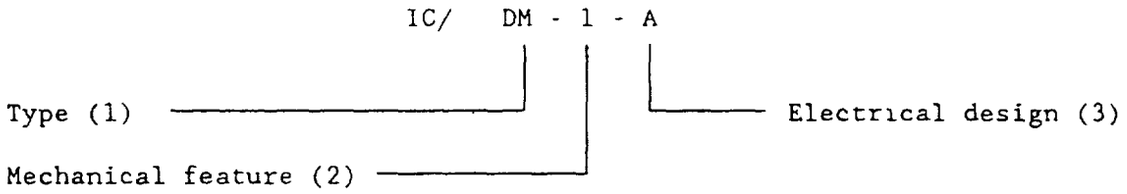
4 Contractor certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices The government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract

5 Regulatory requirements The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical

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6 Preservation, packaging, packing, labeling, and marking The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order

7 Identification numbers The part identification number shall be determined using table 1 and the configuration scheme outlined below



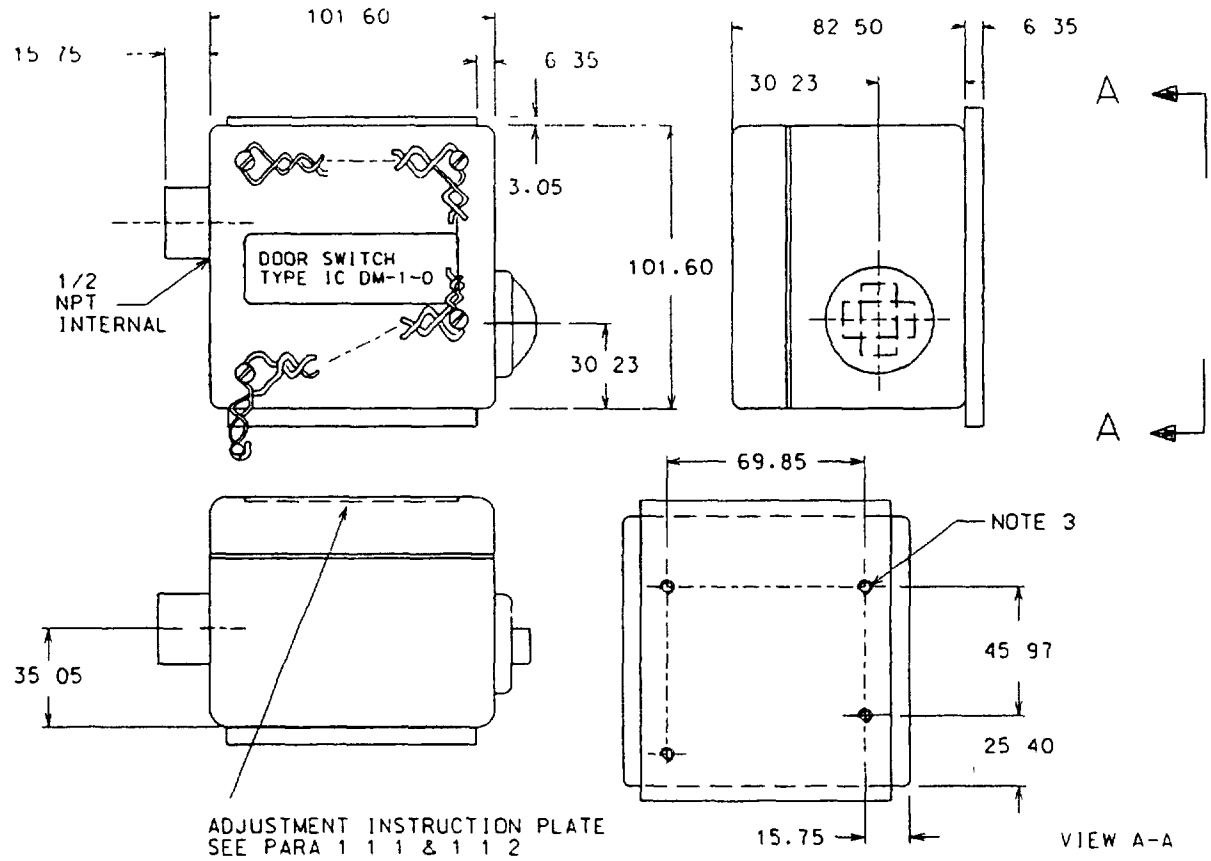
(1) Type of operation
(2) Mechanical feature
(3) Electrical design

DM - Door
1 - Adjustable
A - Characteristics from Row A of table 1

CIVIL AGENCY COORDINATING ACTIVITIES
GSA-FSS

PREPARING ACTIVITY
NAVY - SH
(Project 5930-N753)

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NOTES

1 Unless otherwise indicated, all dimensions are in millimeters
2 Tolerances shall be as follows
  Two place decimals ± 0 02 mm
  Three place decimals ± 0 005 mm
  Angular dimensions ± 30 minutes
3 Base plate shall be secured to the switchbox with M6 3X1 hardware Screws shall be accessible from inside for removal of base plate prior to welding of plate to mounting surface
4 Cover shall be fastened by M6 3X1 drilled hex head bolts 51 mm in length
5 Security wire and seals shall be provided for installation as indicated

FIGURE 1 Switch, Door, Type IC/DM-1-( )

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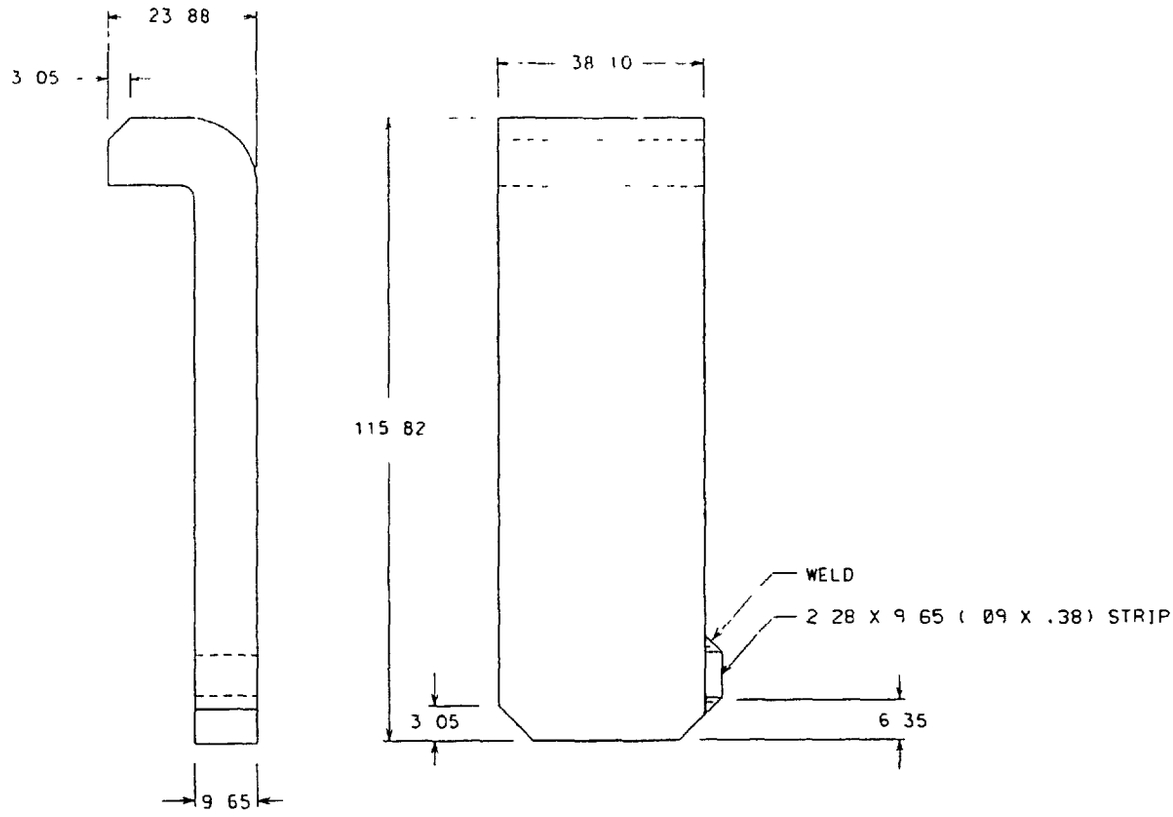


FIGURE 2 Switch Striker

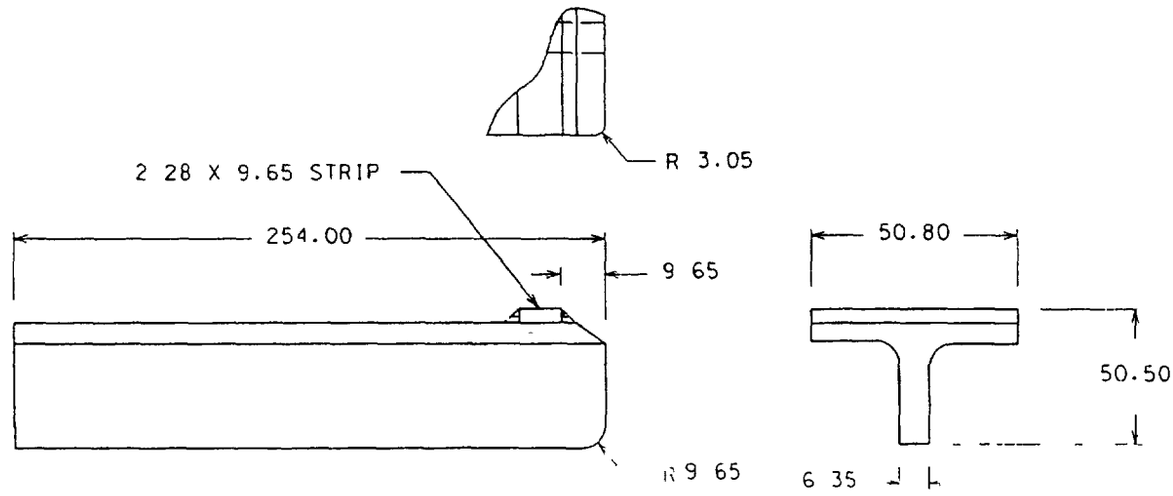


FIGURE 3 Tee Striker

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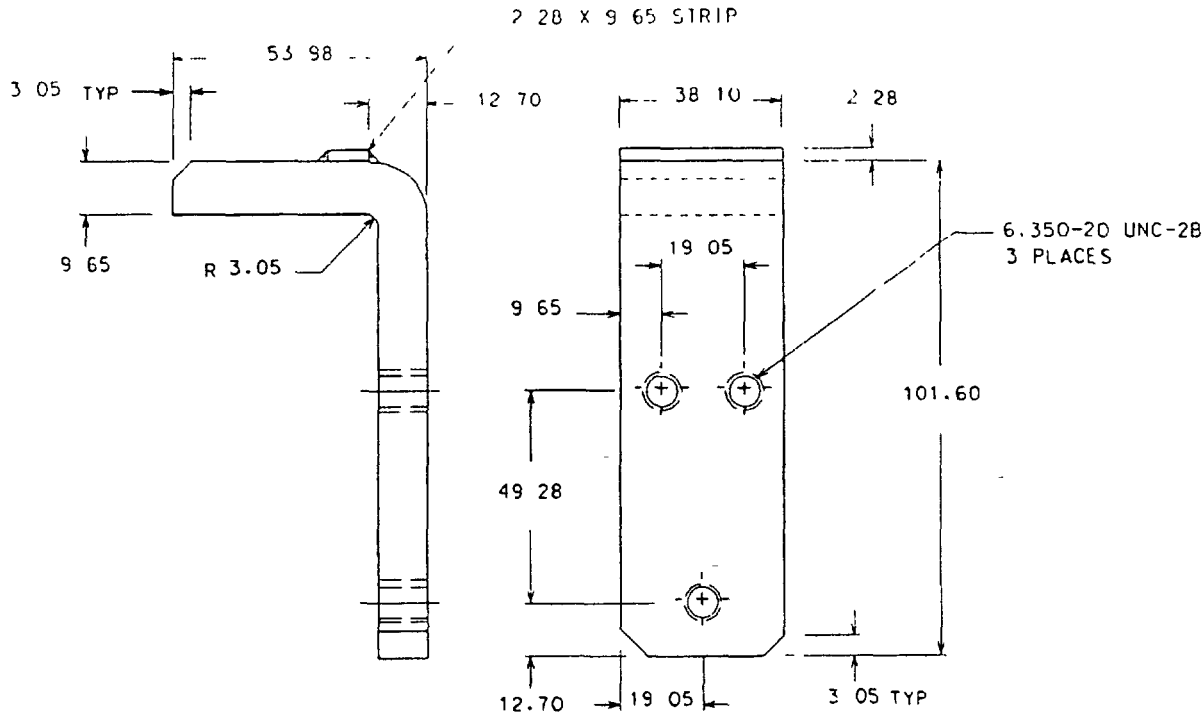


FIGURE 4 Switch Striker

DOOR SWITCH
TYPE IC/DM - 1 - ( )

N S N

NAVAL SEA SYSTEMS COMMAND

MFG PART NO

CONTRACT NO

FIGURE 5 Identifcation Plate

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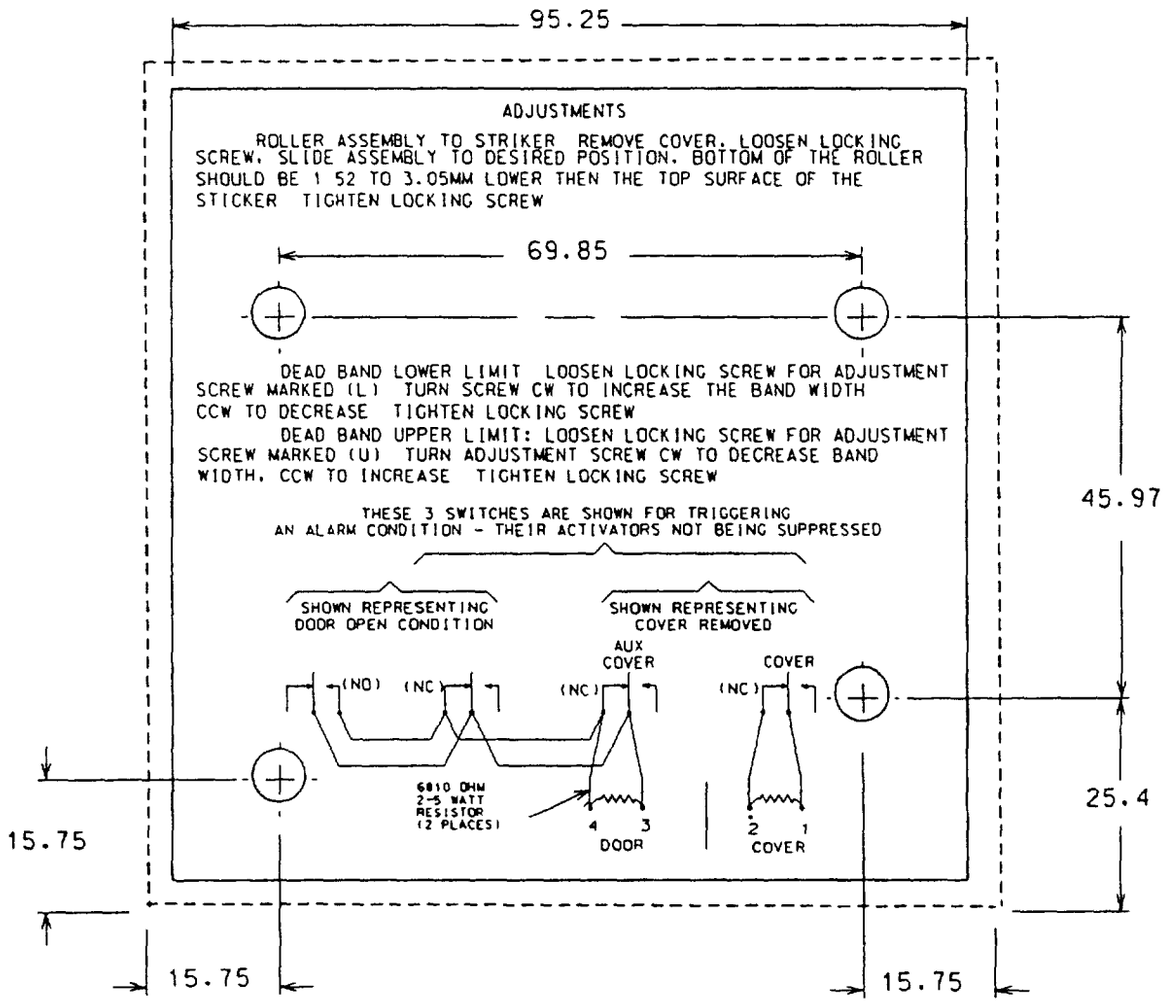


FIGURE 6 Instruction Plate

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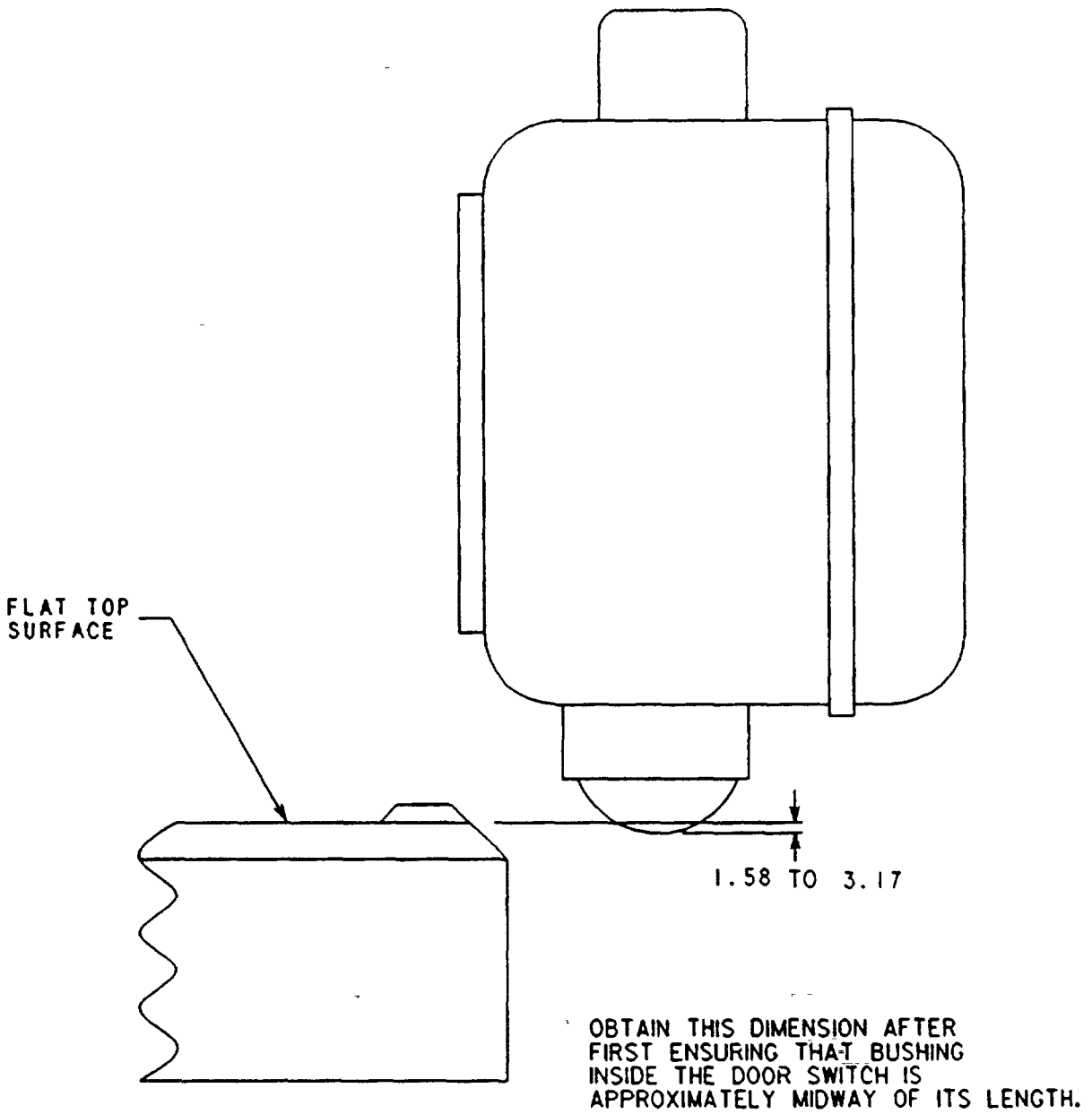


Figure 7 Roller Assembly to Striker Adjustment

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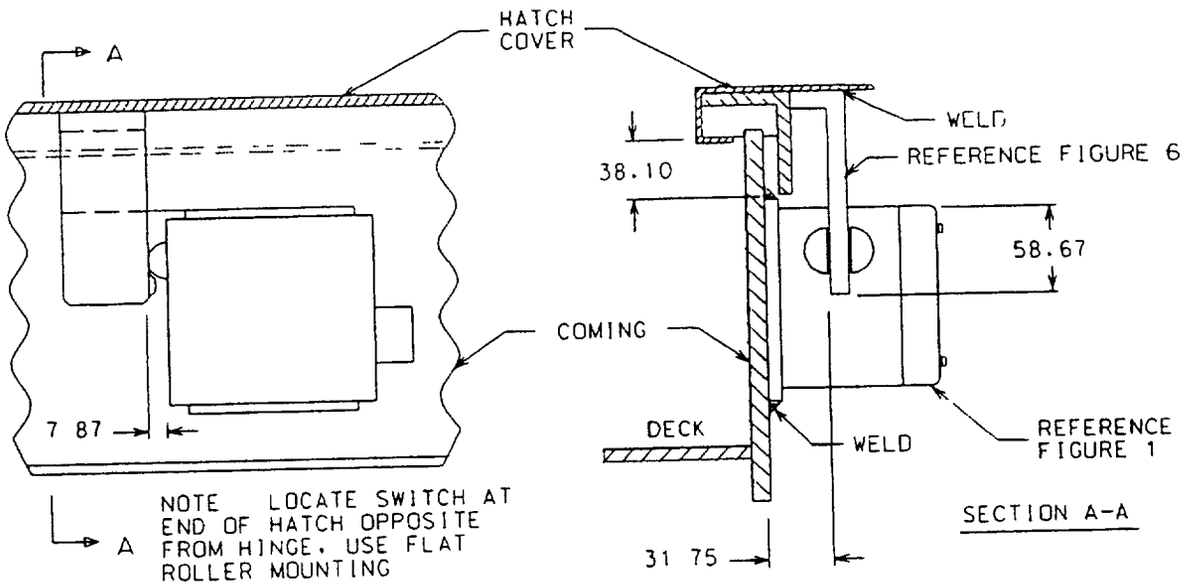


Figure 8 Typical Hatch Installation

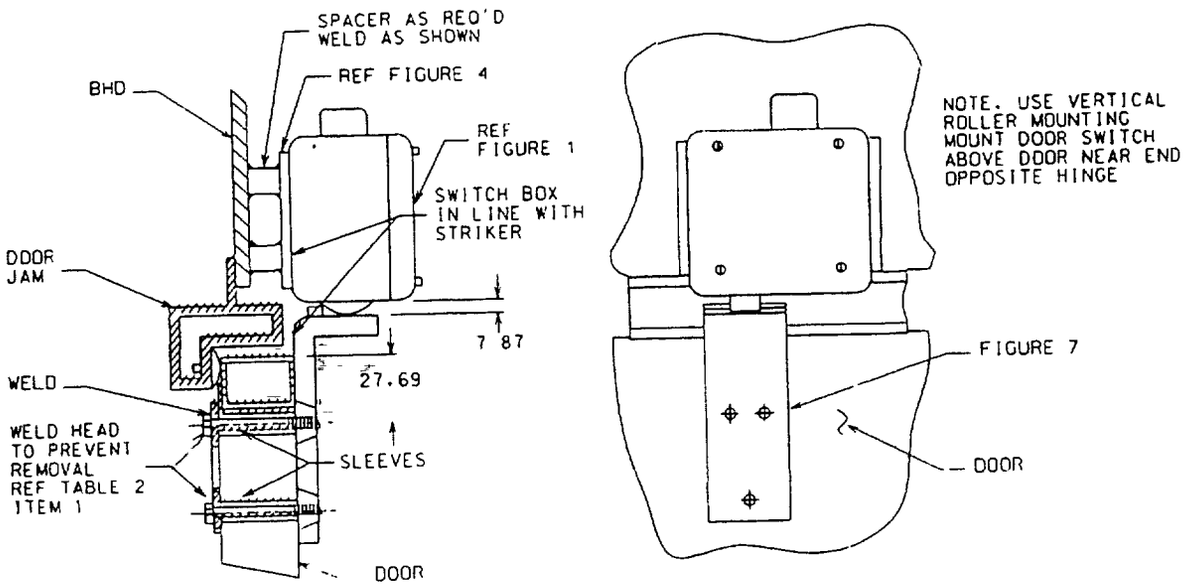


Figure 9 Typical Metal Joiner Door Installation

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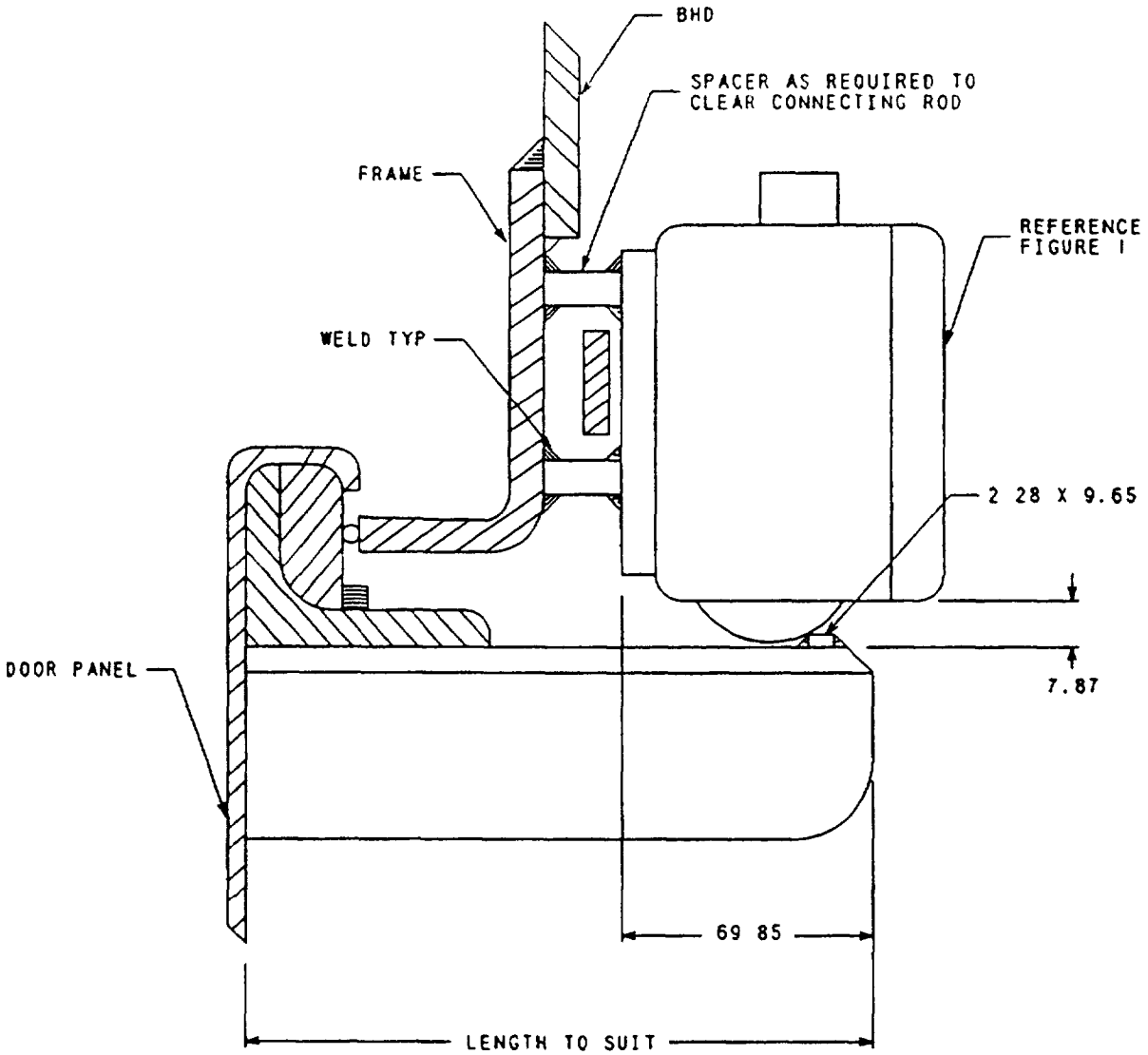


Figure 10 Typical Watertight Door Installation

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the comment number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of this form

NOTE This form may not be used to request copies of documents nor to request waivers or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-50626	2 DOCUMENT DATE (YYMMDD) 940718

3 DOCUMENT TITLE
SWITCH, DOOR TYPE IC/DM-1-()(METRIC)

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

A. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	c. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

A. NAME Technical Point of Contact (TPOC) MR. C.R. TRUDE, SEA 03K32 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN. TPOC: 703-602-8262 8-332-8262
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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